Where OCR Breaks on Bangla Mathematics Textbooks: An Error Analysis of Tesseract, LLM Post-Correction, and Vision LLMs for Retrieval-Augmented Math Tutoring

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Abstract

Retrieval-augmented generation (RAG) tutors depend on clean text extracted from source textbooks, yet the national (NCTB) mathematics textbooks used by millions of secondary students in Bangladesh are distributed as scanned PDFs that mix Bangla script, Latin variables and mathematical notation. We study how optical character recognition (OCR) fails on these books and how much of the damage can be repaired. On 45 pages of a Grade 9–10 chapter on sets and functions (22 Bangla, 23 English), an LLM post-correction pass made 1,024 edits to Tesseract output; 52.6% of them targeted mathematical notation (symbols, exponents and set brackets) rather than ordinary words, and Bangla–Latin digit confusion appeared only in Bangla pages. On three hand-verified Bangla pages, Tesseract had a character error rate (CER) of 11.0% and misread 99% of mathematical symbols; text-only LLM post-correction lowered CER only to 8.4% (symbols: 67% wrong), whereas a vision LLM reading the page image reached 0.24% CER with no symbol errors. [TODO: 1 sentence: effect on RAG answer accuracy for 20 questions (Day 6).] We release our annotated pages and code to support Bangla educational NLP.

1 Introduction

- Problem: AI math tutors for Bangla-medium students need the official NCTB textbooks as grounding; the books exist only as scanned images.
- Why it is hard: mixed scripts (Bangla words, Latin variables, both Bangla and Latin digits) plus math notation ($\in, \cup, \cap, \emptyset, \therefore$, superscripts, set braces).
- Our system context: MathAI Bangladesh, a RAG tutor (Section 3). Observation that motivated this paper: answers were often irrelevant or wrong, and the cause traced back to OCR text.
- Contributions:
  1. A categorised analysis of 1,024 OCR corrections on 45 bilingual textbook pages.
  2. A small character-level ground truth [TODO: N pages] and a CER comparison of three extraction methods.
  3. Evidence that LLM post-correction is a double-edged tool: it fixes notation but also deletes or rewrites correct text.
  4. [TODO: RAG answer accuracy before/after correction.]

2 Related Work

[TODO: Keep short (3 paragraphs). Only cite papers you have actually opened.]

- OCR engines: Tesseract [1]. [TODO: Bangla OCR work; math OCR / formula recognition work.]
- LLM-based OCR post-correction. [TODO: find 2–3 papers.]
- Retrieval-augmented generation [2]; effect of noisy retrieval corpora. [TODO: find 1–2 papers.]
- Bangla NLP resources and educational AI. [TODO: find 1–2 papers.]

3 System and Data

Corpus. Five NCTB mathematics textbooks (Grades 6–10, General and Higher Math), each in a Bangla and an English edition: 10 scanned PDFs, 61 chapters, about 2,562 pages.

Pipeline. Tesseract OCR (`ben+eng`) → chunking (1,000 characters, 200 overlap, sentence-aware) → multilingual-e5-large embeddings → Qdrant vector store (3,907 chunks) → retrieval → answer generation with a multi-LLM cross-checker (two models plus a tie-breaker).

Study subset. Grade 9–10 General Math, Chapter 2 (Sets and Functions): Bangla pages 26–47 and English pages 27–49 (45 pages, 60,521 raw OCR characters).

4 Method

M1: Tesseract (baseline). Raw output, `ben+eng`.

M2: Tesseract + LLM post-correction. Each page's OCR text is sent to a Groq-hosted LLM with a short Bangla prompt asking it to fix only mathematical notation (e.g. `x2`→x^2, `...`→$\therefore$); the edit list (before/after spans) is logged per page. The exact model version used for this run could not be recovered from the project logs; [TODO: state as such or confirm].

M3: Vision LLM transcription. The page image (300 DPI) is transcribed directly by `gemini-3.6-flash` (October 2026, temperature 0) with a prompt asking for an exact, uncorrected transcription with Unicode math symbols.

Error categories. Each logged edit is assigned automatically to one of seven categories (math symbol, superscript/exponent, bracket/set notation, Bangla–Latin digit swap, punctuation/spacing, deletion, other) by rule-based heuristics (code released).

Ground truth and metric. Three Bangla pages (PDF pages 30, 36, 44; printed pages 25, 31, 39) covering definitions and worked examples (30, 44) and an exercise set (36) were verified character by character against the page image by the author, a mathematics teacher. To save effort the verification started from the vision-LLM transcription and corrected it; this may favour M3, so further pages are being verified starting from the Tesseract output instead (Section 6). CER is the Levenshtein distance divided by reference length after NFC normalisation and removal of all

whitespace (a variant keeping single spaces gives the same ranking). We also break errors down by character class.

Downstream RAG test. [TODO: 20 questions from Chapter 2 with reference answers; answers marked correct/incorrect by the author.]

5 Results

5.1 What the LLM corrector changed

	Bangla	English	Total
Pages	22	23	45
Raw OCR characters	29,615	30,906	60,521
LLM edits	557	467	1,024
Edits per page	25.3	20.3	22.8

Table 1: LLM post-correction edits on Chapter 2.

Category	Bangla	English	Total	%
Math symbol	172	182	354	34.6
Other (word / mixed)	194	94	288	28.1
Punctuation / spacing	47	80	127	12.4
Superscript / exponent	46	79	125	12.2
Bracket / set notation	43	16	59	5.8
Bangla–Latin digit swap	43	0	43	4.2
Deletion	12	16	28	2.7
Total	557	467	1,024	100

Table 2: Edits by category (automatic, heuristic labels).

Observations:

- Mathematical notation (symbols, exponents, set brackets) accounts for 52.6% of edits: the dominant OCR failure is notation, not Bangla words.
- Bangla–Latin digit confusion (e.g. `x` read as Bangla five) occurs only on Bangla pages, a mixed-script failure specific to Bangla mathematics books.
- Some edits are harmful: correct text deleted, a written number word rewritten as a digit, plain digits turned into superscripts. On the verified pages, Bangla-letter errors *increased* after correction (0.98%→1.47%; Table 4): the corrector rewrites text it cannot see.

5.2 Character error rate

Tesseract reads Bangla prose well (under 1% error on Bangla letters) but fails almost completely on mathematical notation: 98 of 99 math symbols and a third of the Latin variables were wrong.

Method	p.30	p.36	p.44	All (micro)
M1 Tesseract	7.98	13.43	13.09	11.03
M2 Tesseract + LLM correction	6.53	10.48	9.10	8.40
M3 Vision LLM	0.07	0.11	0.61	0.24

Table 3: CER (%) on three verified Bangla pages (3,273 reference characters). A second, independent Tesseract run gave 10.91%.

Character class	n	M1	M2	M3
Bangla letters / vowel signs	1,836	0.98	1.47	0.11
Bangla digits	34	32.35	32.35	14.71
Latin letters (variables)	296	32.77	18.92	0.00
ASCII digits	291	0.69	2.41	0.00
Math symbols	99	98.99	66.67	0.00
Brackets / punctuation	717	14.92	7.53	0.00

Table 4: Error rate (%) by reference character class.

Text-only correction recovers some notation but cannot restore what OCR has destroyed (e.g. a garbled `(2,%,2]` for $\{x, y, z\}$), and introduces new errors in Bangla text and digits. The vision LLM's few errors were confined to Bangla digits in a vertically printed margin year and one extra vowel sign in a page footer.

5.3 Effect on RAG answers

[TODO: Accuracy on 20 questions with current (uncorrected) index; if time, after re-indexing corrected text (Day 6).]

6 Discussion and Limitations

- Small scale: one chapter, three verified pages, one annotator. Ground truth started from the M3 output, which may favour M3.
- Category labels are heuristic; a manual check of a sample is [TODO: done / future work].
- Models change quickly; results are tied to the model versions named.
- Future work: all 61 chapters, human-reviewed corrections, full RAG evaluation.

7 Conclusion

[TODO: 3–4 sentences after results are final.]

References

[1] R. Smith. An overview of the Tesseract OCR engine. In *Proc. ICDAR*, 2007.

[2] P. Lewis et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in NeurIPS*, 2020.